Freight Rate Prediction

Oct 31, 2026

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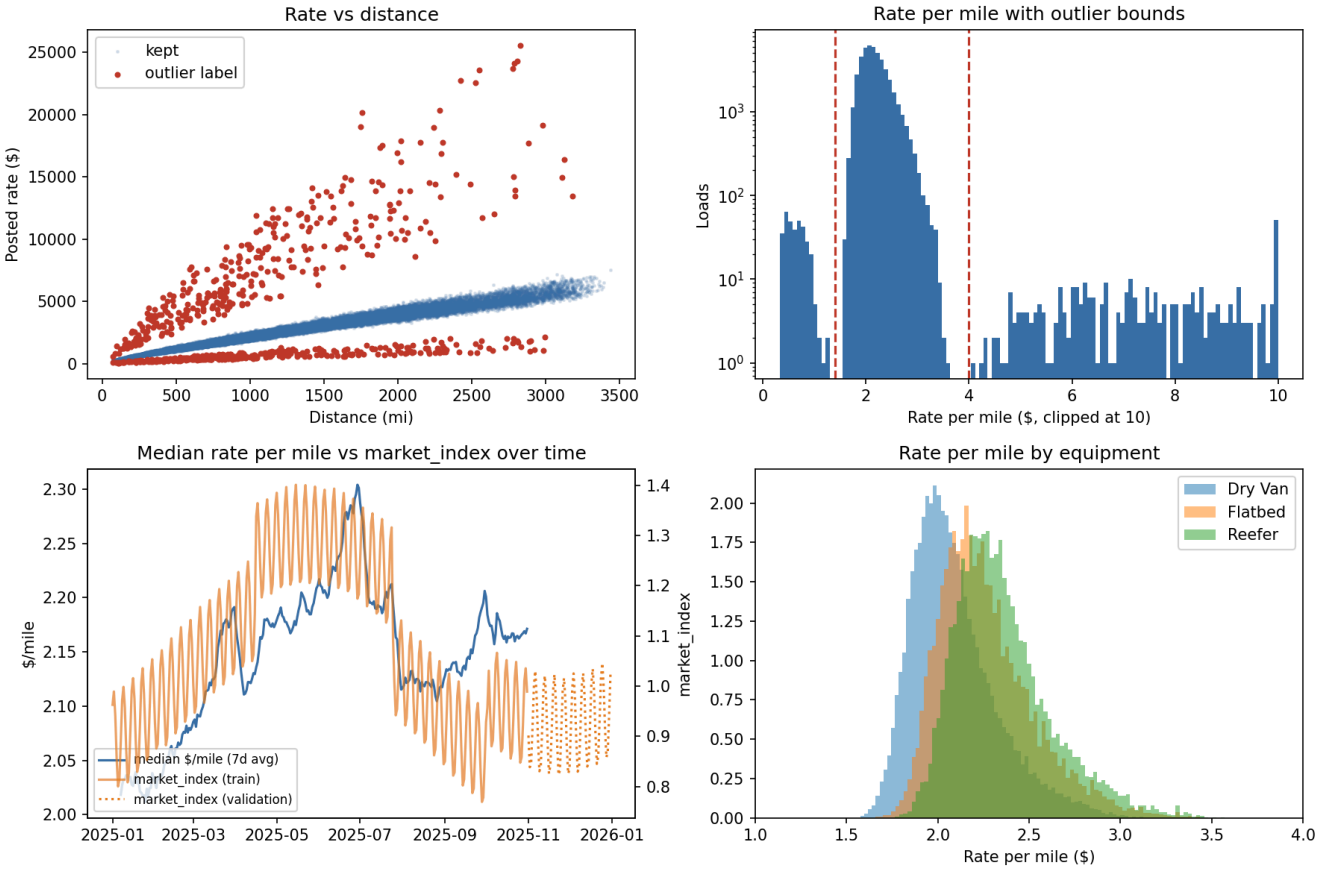
Goal

I built a model that predicts the posted rate of a truckload. It learns from 48,000 labelled loads (Jan–Oct 2025) and predicts 12,000 future loads (Nov–Dec 2025).

The deliverables are a code repository, `validation_predictions.csv` with one predicted rate per load, a report, and a December prediction chart. On the closest test to the real task, the model's average error is $29 per load, against $83 for a simple baseline.

What the data showed

- **Distance drives the price.** A typical load costs about $2.15 per mile. Reefer and Flatbed cost more than Dry Van, and short trips cost more per mile.
- **Market index moves prices day to day.** It is one value per day for the whole market. A higher index means higher rates on that day.
- **Prices follow a quarterly cycle.** They stay flat for about 7 weeks, then rise about 5% into the end of each quarter, and reset. Each quarter also starts about 2% higher than the last. December is a quarter end, so December prices should climb.
- **8 cities are new.** Chicago, Charlotte, San Diego and 5 others appear only in the prediction data, on 1,447 loads. The model therefore uses map position and distance, never city names.



Top: bad labels (red) sit far from the main band, and the dashed lines mark the cut-offs. Bottom left: the base rate climbs toward each quarter end. Bottom right: Reefer and Flatbed cost more per mile than Dry Van.

How I cleaned the data

Problem	Rows (labelled data)	Fix
Negative weight	292	Sign error, so made positive
Missing weight	300	Left blank and flagged; the model handles blanks
Missing market index	374	Filled with that day's median, since it is one value per day
Impossible prices (under $1.4 or over $4.0 per mile)	677	Treated as wrong labels and removed from training

The same weight and market-index fixes are applied to the 12,000 loads being predicted.

How I tested it

I split the data by time, never at random, because the real task is predicting the future. A random split would let the model peek at each month's market level and look better than it is.

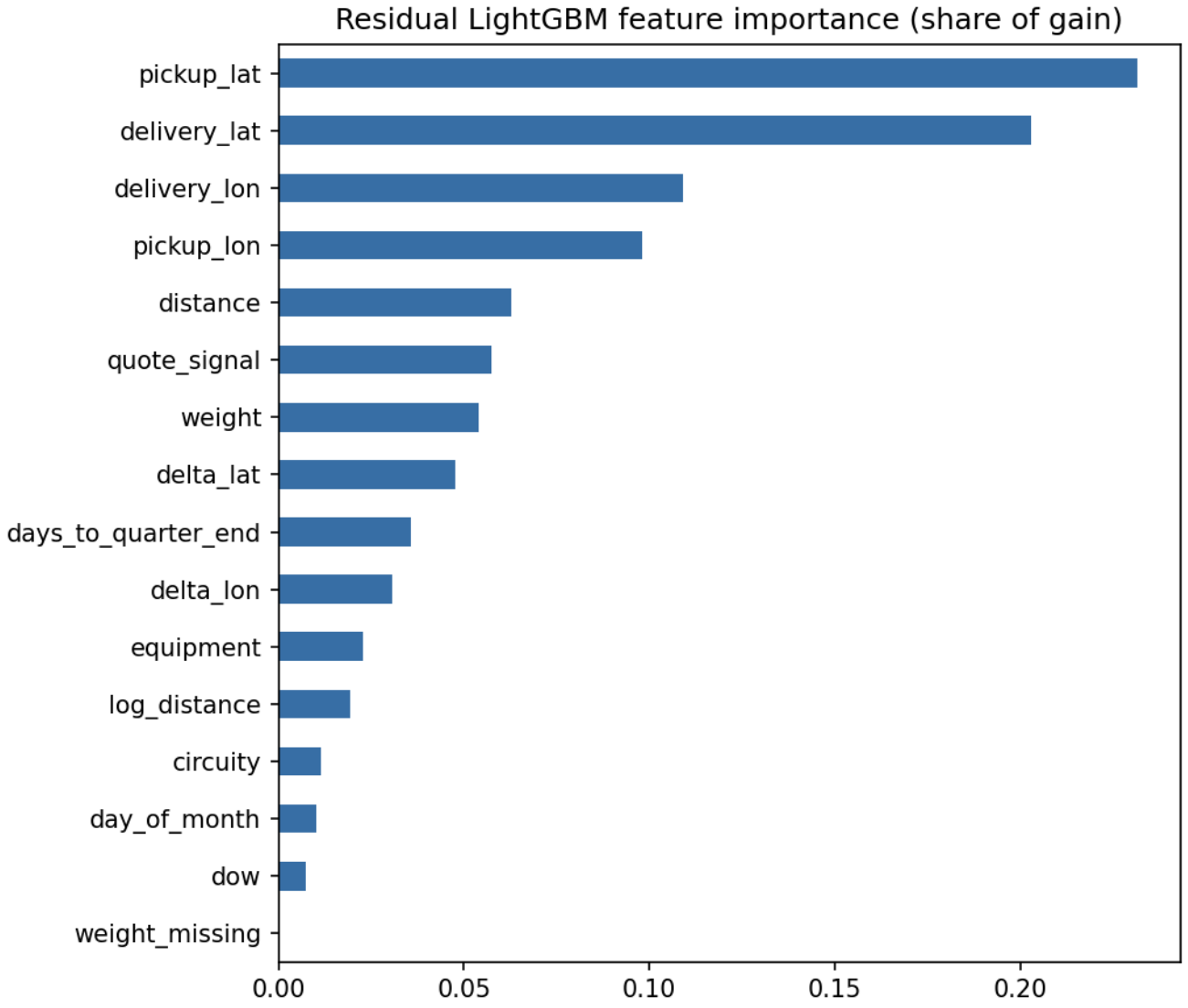
1. **Six time folds.** Each fold trains on all earlier months and tests on the next two months, e.g. train Jan–Aug, test Sep–Oct.
2. **Main holdout.** Sep–Oct is the main test, because it is closest to the real Nov–Dec task.
3. **New-city test.** In each fold, 8 random cities are hidden from training and the model is scored on loads touching them. This copies the 8 new cities in the real prediction data.
4. **Final model.** After choosing the model, I retrained it on all clean Jan–Oct data.

The model

The model predicts price per mile, then multiplies by distance. It works in two stages:

1. **A simple linear model** captures the trends that keep going into the future: the market index, the quarter-to-quarter rise, and the ramp into quarter end.
2. **LightGBM** (a tree-based model) corrects what the first stage misses: route geography, equipment, weight and day-of-week effects.

I used two stages because tree models alone cannot extend a rising trend into months they have never seen. In testing, LightGBM alone had about twice the error of the two-stage model.



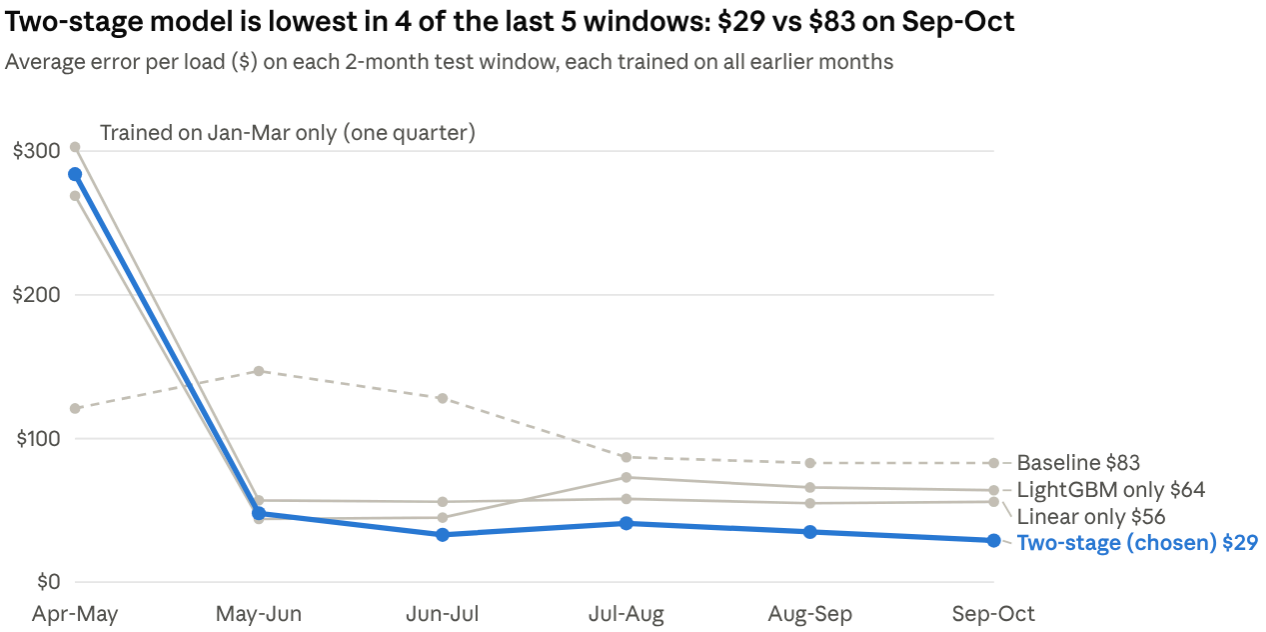
In the LightGBM stage, route position (pickup and delivery latitude and longitude) matters most. Market index and the quarter trend are handled by the linear stage, so they do not appear here.

Results

The two-stage model is the most accurate, including on the hidden cities.

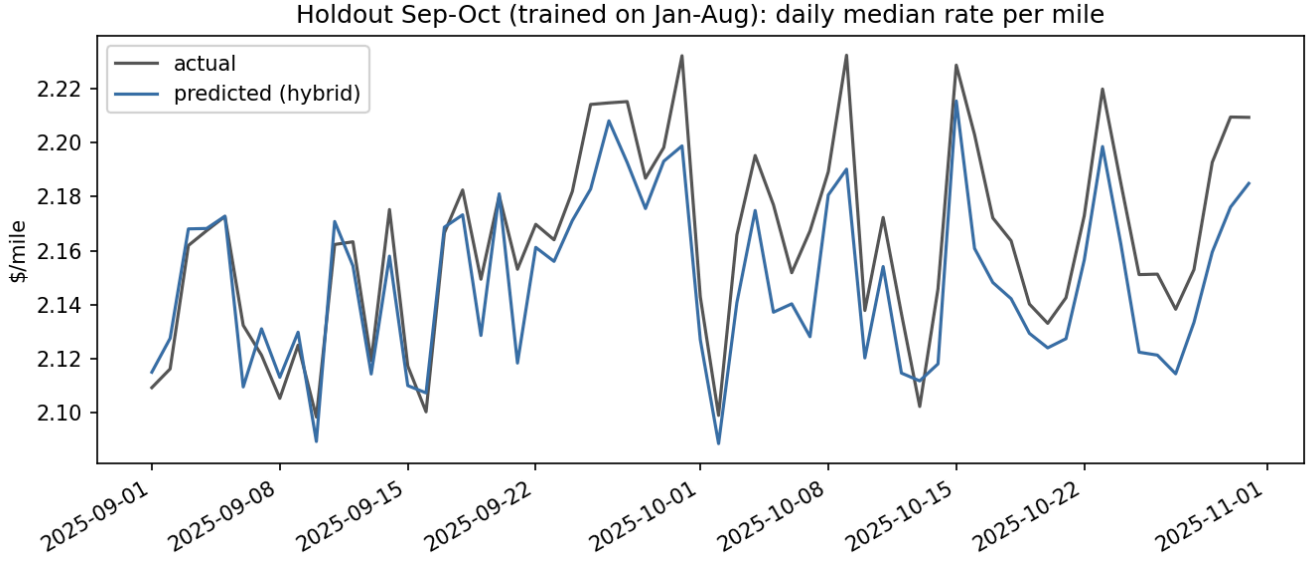
Model	Average error, Sep–Oct test	Average error, hidden cities
Two-stage (chosen)	$29	$33
Linear only	$56	$51
LightGBM only	$64	$63

Model	Average error, Sep–Oct test	Average error, hidden cities
Baseline (median $/mile lookup)	$83	$78



outputs/cv_results.csv · 6 time-ordered test windows, Apr–Oct 2025

One weak spot: the earliest fold, trained on only Jan–Mar, does worse than the baseline. With one quarter of data the model cannot separate the market effect from the trend. The final model trains on four quarters.



On the Sep–Oct holdout, the predicted daily rate follows the actual one, including the climb into the end of September and the reset in October.

The December chart (Lexington to Fort Wayne, 360 miles, Dry Van) predicts about $820 to $844. The price rises through the month into the quarter end, with a small weekly ripple from the market index.

How to run it

Install the packages with `pip install -r requirements.txt`, then run `sh run_all.sh`. It runs four scripts in order:

1. `src/eda.py`: exploration charts and the data-quality summary
2. `src/train.py`: time-based testing, model comparison and the final model
3. `src/predict.py`: writes `validation_predictions.csv` and the December chart
4. `src/make_report.py`: builds `report/report.docx`

The cleaning steps and both model stages live in `src/pipeline.py`.